

Harish Prabhu P

AI & Automation Engineer | Production ML, GenAI & Delivery

AI engineer who takes a business problem from the first conversation to something running in production with a number against it. Sat as **Project-In-Charge inside four FairPrice Group business units** and as **point of contact for external stakeholders**, turning operational problems into specifications and then into working software. Shipped **production machine learning**, not prototypes: a **BERT and BERTopic pipeline** on **BigQuery** classifying **50,000+ monthly records in real time**, cutting manual triage **40%**. Defines the success measure before building and evaluates against it afterwards, **GenAI and LLM** work included. Builds outside work too, across **full-stack** and personal **AI and LLM** projects. Reading a BTech in Computing (**Data Science and AI**) at NTU under a work-study programme.

Work Experience

Executive - Service Analytics (Customer Service) | FairPrice Group, Singapore

November 2025 - May 2026

- **Built and deployed a production NLP classification pipeline** using **BERT models (BERTopic, transformer embeddings)** and **BigQuery**, categorising **50,000+ monthly interactions in real time** to enable automated routing and reducing manual triage by **40%**
- **Architected an automated response system** on **Google Apps Script, REST APIs and webhook integrations**, processing **1,000+ daily transactions** through intelligent routing, and improved service level from **75% to 98%** while removing **20 person-hours of manual work daily**
- **Worked embedded with four business units** (Retail Stores, Ecommerce, Food Services, Link) as **Project-In-Charge** and as **point of contact for external stakeholders**, framing problems and translating them into **technical specifications, user stories and acceptance criteria**
- **Applied GenAI and LLM prompt engineering to evaluate automation impact** against customer experience metrics, defining **operational ROI measurement frameworks** before build so results could be verified rather than assumed
- **Owned delivery across the software lifecycle**: design, build, **test plans and test cases, UAT coordination**, defect tracking in **JIRA and Confluence**, and release with post-launch measurement
- **Mapped the microservices and API call flow** to integrate the **Beverage Container Return Scheme (BCRS)**, a national regulatory programme, into an existing retail and ecommerce platform
- **Grounded decisions in evidence**, running **root cause and funnel diagnostics** over **50,000+ monthly records** and building the **BigQuery and Tableau** dashboards business units adopted

Cage Cashier (High-Value Transactions, AML/CDD) | Marina Bay Sands, Singapore

November 2024 - October 2025

- Worked to exacting standards in a licensed, regulated environment, reconciling **S\$500,000+ in daily transactions at 100% accuracy** and troubleshooting system discrepancies under time pressure

Customer Care Officer (Data & Reporting) | Tele-Centre Pte Ltd, Singapore

February 2020 - August 2020

- Ranked **1st in performance** handling **10,000+ time-critical cases at 98% accuracy** on MOH Emergency and 993 Ambulance Hotline operations, and **trained new staff** to a **25% efficiency lift**

Educational Background

Bachelor of Technology in Computing (Work-Study Degree)

Nanyang Technological University, Singapore

August 2025 - July 2029

- Specialization In Data Science and Artificial Intelligence
- Built and shipped full-stack projects (Python, FastAPI, React, Node.js, MongoDB), with a further project in progress

India Standard 12 (Eq. A Levels)

Aristo Public School, Tamil Nadu, India

June 2020 - June 2022

Contact

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Skills

AI & Machine Learning

- Production ML Deployment & Monitoring
- NLP & Text Classification (BERT, BERTopic)
- GenAI & LLM Prompt Engineering
- LLM Evaluation Against Business Metrics
- RAG System Development
- Feature Engineering & Real-Time Processing

Software Engineering

- Full Software Development Lifecycle
- API & System Integration Design
- Microservices & Data Flow Mapping
- Test Planning, UAT & Defect Tracking
- Data Pipeline & ETL Engineering
- Full-Stack Application Development

Delivery & Analysis

- Business Problem Framing & Translation
- Stakeholder & External Client Engagement
- Statistical Analysis & A/B Testing
- Root Cause & Funnel Diagnostics
- Agile Kanban Delivery
- Impact Measurement & ROI Frameworks

Tech Stack

Languages & Frameworks

- Python (Pandas, NumPy, scikit-learn, Pydantic, pytest)
- SQL (Expert), FastAPI, React, Node.js, TypeScript
- Java, MongoDB (knowledge level)

AI, Data & Cloud

- BERT, BERTopic, Transformer Embeddings
- Google Vertex AI, Gemini; LangChain, LlamaIndex, MCP
- Ollama, vLLM, LM Studio (self-hosted inference)
- GCP: BigQuery, Apigee; Docker, Git, CI/CD
- Azure, AWS (NTU projects); Airflow, Prefect

Certification

- Certified Full Stack Developer (NTU)
- Large Language Models (Google Cloud)
- Google AI Essentials
- Data Science Math Skills (Duke)
- Data Science Methodology (IBM)

Additional Information

- **Citizenship**: Singaporean
- **Availability**: Weekdays full-time through 2029, incl. 12-month NTU industry attachment from Aug 2028; Saturdays for studies
- **Portfolio**: harishprabhu-portfolio.vercel.app